DAA-2: Job Sequencing with Deadlines

Technique: Greedy Method

1. Introduction

Job Sequencing with Deadlines is a scheduling problem in which each job takes one unit of time and has a deadline and a profit. The objective is to schedule jobs so that every selected job finishes on or before its deadline and the total profit is maximum.

2. Greedy Approach

The Greedy method selects the job with the highest profit first. Each selected job is placed in the latest available time slot on or before its deadline. Placing a job as late as possible keeps earlier slots free for jobs with smaller deadlines.

3. Algorithm / Steps

1. Accept the number of jobs and their job ID, deadline, and profit.

2. Sort all jobs in descending order of profit.

3. Find the maximum deadline and create that many empty time slots.

4. For each job, search backward from its deadline for the latest free slot.

5. If a free slot is found, schedule the job and add its profit.

6. Display the scheduled jobs and the maximum total profit.

4. Example with Calculation

Consider the following three jobs. Each job requires one time slot.

Job	Deadline	Profit
J1	2	100
J2	1	19
J3	2	27

Step 1: Sort jobs by profit in descending order.

Order	Job	Deadline	Profit
1	J1	2	100
2	J3	2	27
3	J2	1	19

Step 2: Maximum deadline = 2, so create two slots: [Empty, Empty].

Step 3: Schedule each job
• J1 (deadline 2, profit 100): slot 2 is free, so place J1 in slot 2. Slots: [Empty, J1].
• J3 (deadline 2, profit 27): slot 2 is occupied; slot 1 is free, so place J3 in slot 1. Slots: [J3, J1].
• J2 (deadline 1, profit 19): slot 1 is occupied, so J2 cannot be scheduled.

Final schedule: Slot 1 = J3, Slot 2 = J1

Total profit calculation:
Total Profit = Profit(J3) + Profit(J1)
= 27 + 100
= 127

Result: Scheduled Jobs = [J3, J1] and Maximum Profit = 127.

5. Time and Space Complexity

Operation	Complexity
Sorting jobs	O(n log n)
Finding slots	O(n × D), where D is maximum deadline
Overall time	O(n log n + nD)
Space	O(D) for the time-slot array

6. Applications

• CPU and task scheduling
• Project and resource scheduling
• Production planning
• Assigning jobs to limited time slots

7. Conclusion

Job Sequencing with Deadlines uses the Greedy technique to maximize total profit. By selecting higher-profit jobs first and placing each in the latest available slot before its deadline, an optimal schedule is obtained for the standard unit-time job sequencing problem.